Spheroid, basement membrane, matrix:

the three-entity model and the arrows between them

the container note · `prototype/ecm` · `log/okuda_ECM` · companions: `note_junction`, `note_fibre`, `note_surface`, `note_sheet` · 10 August 2026

1. Mechanism

An epithelial spheroid growing inside an extracellular matrix is *three* entities and not two, and the middle one is not a passive spacer. The literature fixes its proportions, and they are not the convenient ones.

The basement membrane is thin, stiff and squat-anchored. It is $\sim$100 nm thick; its adhesions are plaques $\sim$350 nm across on $\sim$700 nm centres joined to the cell across a $\sim$30 nm gap [11], so an adhesion is three times *wider* than the sheet is thick and a third as long; and the sheet is 10 to 100 times stiffer than the stroma outside it, not equal to it.

It is not a permanent solid. Collagen IV in the fly embryo has a half-life of 7–10 hours [10] and in mouse hair follicle about 3, and slowing that turnover disrupts morphogenesis. A sheet enclosing a tissue that triples its radius does not stretch to 2.4; it is *replaced* at the new size while it is being stretched.

And it is the interface. While the sheet is intact the epithelium does not touch the stroma: load goes cell $\to$ plaque $\to$ sheet $\to$ anchoring fibril $\to$ stroma [12]. An epithelium that *also* pushes the stroma directly is counting the same push twice. That single observation is what removes `cell_to_ecm` from the model and puts two edge sets in its place.

2. Equations and symbols

Each equation below belongs to one arrow of Fig. 1b. The companion notes carry the mechanics of each entity; this note carries what passes *between* them.

Mass, not just mechanics. The sheet carries an areal density ρ of collagen IV produced by the epithelium and removed by proteolysis, on a surface whose area is growing:

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} = \underbrace{s(\mathbf{x},t)}_{\text{secretion}} - \underbrace{\frac{\rho}{\tau_{\mathrm{bm}}}}_{\text{turnover}} - \underbrace{\rho\frac{\dot{A}}{A}}_{\text{dilution}}, \qquad s^* = \rho\Big(\frac{1}{\tau_{\mathrm{bm}}} + \frac{\dot{A}}{A}\Big), \qquad \tau_{\mathrm{bm}} = \frac{t_{1/2}}{\ln 2} \approx 4\text{–}14\ \mathrm{h}. \tag{1}$$

The two removal terms are the same size here — a spheroid tripling its radius over a day has $\dot{A}/A \sim 10^{-1}\,\mathrm{h}^{-1}$ against $1/\tau_{\mathrm{bm}} \sim 10^{-1}\,\mathrm{h}^{-1}$. Turnover is not a correction to growth; it is the same size as growth, which is why the sheet can accommodate it at all.

A nonlinear sheet. A single Young's modulus is wrong by construction: the same membrane reads $\sim$0.4 MPa at low strain and $\sim$3 MPa at high [9], and a growing spheroid takes it across that range:

$$E(\lambda) = E_0\big[1 + \beta(\lambda - 1)\big], \qquad \lambda = \sigma_{\max}(\mathbf{F}), \qquad \frac{E_{\mathrm{bm}}}{E_{\mathrm{ecm}}} \in [10, 100]. \tag{2}$$

MPa is not a quantity this box has — it is dimensionless and the stroma's modulus is the number 15 — so the falsifiable forms are the two ratios and never the pascals.

Adhesion, with its reaction and with slip. A plaque is two-sided: whatever force it applies to the sheet it applies equally and oppositely to the cell, which a replayed epithelium discards. And the sheet is not pinned — it *slides* — so the tangential law is friction against relative velocity and not a spring to a frozen direction:

$$\mathbf{f}_p^{\mathrm{adh}} = \kappa_n(\ell_p - \ell_0)\hat{\mathbf{n}}_p - \xi\big(\mathbf{v}_{\mathrm{bm}} - \mathbf{v}_{\mathrm{epi}}\big)_{\parallel}, \qquad \mathbf{f}_p^{\mathrm{epi}} = -\mathbf{f}_p^{\mathrm{adh}}, \tag{3}$$

$$n_p = \Sigma^{-2}\ \text{plaques per unit area}, \quad \Sigma \approx 7T, \qquad \ell_0 \approx 0.3\,T. \tag{4}$$

ℓ_0 is a material property and not a tuning constant: it is what sets the standoff.

Sheet to stroma. Anchoring fibrils [12] load the stroma *through* the sheet, which is also what lets a stiff sheet spread a local epithelial push over a soft matrix:

$$\mathbf{t} = k_a\big(\mathbf{x}_{\mathrm{ecm}} - \mathbf{x}_{\mathrm{bm}}\big)\ \text{on anchored points}, \qquad \mathbf{f}^{\mathrm{bm}} = -\sum_{\text{anchors}} \mathbf{t}. \tag{5}$$

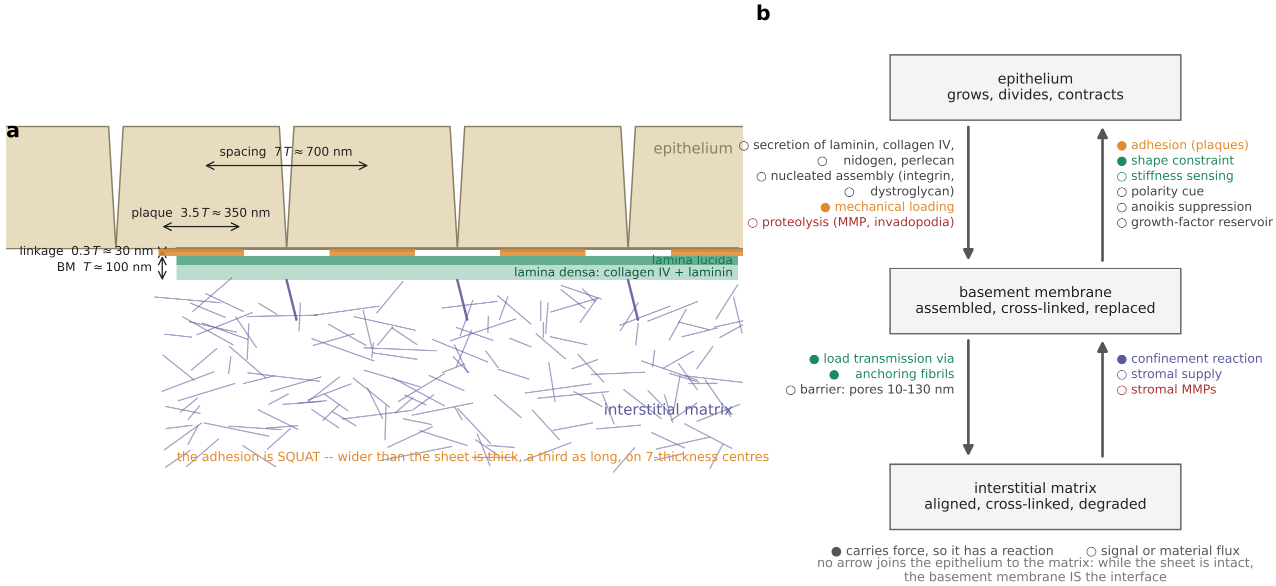

Figure 1: (a) The three entities in section, at the proportions the literature measures rather than the ones that are convenient: the adhesion is a squat plaque, not a thread, and the sheet is thin against its own spacing. (b) Every operator, in the direction it acts. Filled bullets carry a force and must therefore have a reaction; open bullets carry a signal or a material flux. There is no arrow between the epithelium and the matrix.

The barrier is geometric rather than mechanical: pores of 10–130 nm pass solutes, stop cells [8], and are what a proteolytic operator has to open before anything crosses.

The matrix pushes back on the cell cycle. The force the stroma returns, collected by direction and divided by the area each direction covers, is a pressure; a cell facing a stressed matrix grows more slowly, which is what a spheroid grown under a measured external stress does [6, 7]:

$$g = f + \frac{1-f}{1+(P/p_{1/2})^n}, \qquad s_{\text{now}} \leftarrow s_{\text{prev}}\big(1 + g\,(f_{\text{grow}} - 1)\big). \tag{6}$$

Because it is the growth *rate* that is slowed and not the size directly, a small difference compounds over hundreds of frames — which is how a modest difference in stress between one direction and another becomes a difference in shape.

symbol	is	of what
ρ, τ_{bm}, s	areal density, turnover time, secretion	of collagen IV on the sheet, Eq. (1)
A, $\dot{A}/A$	sheet area, its relative growth rate	the dilution term; the SAME size as turnover
E_0, β, λ	low-strain modulus, stiffening, principal stretch	of the sheet, Eq. (2); report RATIOS, never MPa
T, Σ, ℓ_0	thickness, plaque spacing, linkage rest length	$\Sigma \approx 7T$, $\ell_0 \approx 0.3T$; T never appears as geometry
κ_n, ξ	plaque normal stiffness, tangential friction	against RELATIVE velocity — ξ is what lets the sheet stay attached while the tissue slides
k_a	fibril stiffness	sheet to stroma, Eq. (5)
P, $p_{1/2}$, n, f	matrix pressure, gate half-point, sharpness, floor	Eq. (6); $f \le g \le 1$ multiplies the growth RATE

3. The Plexus2 container

Two structural claims, and both differ from what the archived line built. An adhesion is not a body and a fibril is not a particle: each is a *relation between two entities*, which in Plexus2 is an edge set carrying `.pre` and `.post` index buffers rather than a set of nodes with positions. Writing the hemidesmosome as an edge set is what makes its reaction automatic — one operator returns a delta to both endpoints, so the force the sheet feels and the force the cell feels cannot get out of step. The archived line made the adhesion first a field on the membrane and then a column of MPM particles, and both lost the reaction.

set	kind	one element is	state it carries
`spheroid`	node, depth 0	the organ	centroid, cell count
`cell`	node, depth 1	one epithelial cell	A, P, V and targets, age, ρ_f
`vertex`	node, depth 1	a corner of the apical mesh	`pos`
`junction`	**edge** $\subseteq$ `vertex`^{2}	a cell–cell contact	n_e, ℓ_e
`bm`	node on a surface mesh	a patch of membrane	`pos`, $\mathbf{F}$, ρ, age, `occ`
`bm_face`	**hyperedge** $\subseteq$ `bm`^{3}	a triangle of membrane	$\mathbf{D}_m^{-1}$, $\mathbf{C}_0$, A^0, Y_2
`plaque`	**edge** `cell`$\to$`bm`	one hemidesmosome	ℓ_0, load, bound
`fibril`	**edge** `bm`$\to$`mpm_particle`	one anchoring fibril	rest length, load
`mpm_particle`	node, depth 1	a patch of *stroma only*	`pos`, $\mathbf{F}$, fibre direction
`mpm_grid`	**field**	the stroma's momentum	mass, momentum
`protease`	**field**	MMP concentration	scalar, diffusing

operator	kind	acts on	mechanism	whe
epithelium				
`grow_3d`, `divide_3d`	Divide	`cell`	volume growth and cytokinesis [3, 4]	`00;`
`shape_energy_3d`	Lateral	`junction`	vertex-model force balance [5]	`00`
`medioapical_myosin`, `junction_myosin`, `cytokinetic_ring`	Lateral, Structural	`cell`, `junction`	two myosin pools and the ring	`note`
epithelium $\to$ *basement membrane*				
`bm_secrete`, `bm_degrade`	Divide, Die	`bm_face`	Eq. (1) [10, 13]	`note` G19
`bm_stiffness`, `bm_elastic`	Lateral	`bm_face`	Eq. (2)	`note`
basement membrane $\leftrightarrow$ *epithelium*				
`plaque_seed`, `plaque_pull`, `plaque_rupture`	Rewire, Lateral	`plaque`	Eqs. (3)–(4), *with the reaction* [11]	`note`
basement membrane $\leftrightarrow$ *matrix*				
`fibril_seed`, `fibril_pull`	Rewire, Lateral	`fibril`	Eq. (5) [12]	oper
`mesh_contact`	Lateral	`contact`	particle-to-surface, both endpoints in one call	`note`
`bm_barrier`	Exchange	`bm_face`, fields	pores pass solutes, stop cells [8]	oper
matrix				
`seed_ecm`, `mpm_*`, `ecm_stress`	Seed, Exchange, Lateral	`mpm_particle`	MLS-MPM [1, 2]	`note`
`ecm_growth_gate_3d`	Lateral	`cell`	Eq. (6) [6, 7]	`58;`

One schedule, split at exactly one place. Anything carrying a force runs inside the substep block, at the stroma's rate; anything changing topology or mass runs once per frame, because a set whose membership changed mid-substep would be integrated against two different index sets in one step:

$$\begin{aligned}
\text{frame } (\Delta t = 1): \quad & \texttt{cell_geometry_3d} \to \texttt{grow_3d} \to \texttt{bm_growth_gate} \to \texttt{medioapical_myosin} \to \texttt{junction_myosin} \\
& \to \texttt{shape_energy_3d} \to \texttt{reconnect_t1_3d} \to \texttt{divide_3d} \to \texttt{cytokinetic_ring} \to \texttt{junction_myo} \\
& \to \texttt{bm_refine} \to \texttt{bm_secrete} \to \texttt{bm_degrade} \to \texttt{plaque_seed} \to \texttt{plaque_rupture} \\
\text{substep } (\Delta t_{\text{sub}} = \Delta t/8): \quad & \texttt{bm_stiffness} \to \texttt{bm_elastic} \to \texttt{plaque_pull} \to \texttt{fibril_pull} \\
& \to \texttt{mpm_strain} \to \texttt{mpm_scatter} \to \texttt{mpm_grid_update} \to \texttt{mpm_gather}
\end{aligned}$$

Four rules, each buying something specific. *The membrane never enters the grid* — it is ~ 100 nm on a $\sim 200\,\mu$m spheroid, so resolving it as MPM material needs a 3500^3 lattice; as a surface mesh it costs what the epithelium costs and Δx is set by the stroma alone. *The epithelium stays out of the substep block* — it is overdamped and massless, so it does not need the matrix's CFL step, and putting its relaxation inside would multiply its cost by eight and change nothing. *The sheet is massless too*, so its stiffness costs drag rather than substeps: the bound is $\Delta t\,(zk_b + \kappa)/\gamma < 2$, a rate and not a wave speed, and the 10–100 stiffness ratio is paid for with γ. *Particles go on the shell that carries stress, not on the box* — the far field is quiescent, and filling the box at a usable density costs a million particles.

And one thing this deletes. `cell_to_ecm` and `cell_exclude_3d` go, because while the sheet is intact the basement membrane *is* the interface (§1). MPM fibres also stop being asked to act as ropes, which `note_fibre` measured they cannot.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. The per-entity gates live in the companion notes and are indexed in §3; this table is the *system* level — what has to be true of the assembled model, and which arrows are not yet built.

	gate	threshold	measured
the epithelium is unchanged by everything attached to it (`00` is the reference)			
B1	cells, radius, aspect, T1 rate of the bare tissue	the numbers every later run must reproduce	200 → 6076, ×**3.56**, **1.021**, **0.0089**/cell/frame
B2	adding per-junction myosin at $m_e \equiv 1$ changes nothing	bit-identical	**bit-identical**
each arrow carries its reaction			
B3	epithelium → matrix, momentum residual	$< 10^{-6}$	**1.2e-7** max, **1.5e-8** median (`note_surface` S5)
B4	cell ↔ sheet, as bookkeeping and as motion	$< 10\varepsilon_{64}$; drift $< 10^{-12}$	**1.4e-16**, **4.4e-15** (`note_sheet` G9, G10)
B5	sheet → stroma: the stroma is loaded THROUGH the sheet	displacement with the sheet present vs absent	**not built** — `fibril_pull` is the missing arrow
the middle entity does what a membrane does			
B6	the sheet reports its own deformation	> 0.95 of the geometric stretch	**0.9935** as a surface; **0.13** as MPM material (run 130)
B7	ρ held flat while A triples	within 10%	1/11.1 with no secretion (`note_sheet` G24)
B8	the standoff is the linkage's rest length	$\ell/\ell_0 \in [0.9, 1.1]$	**3.43** (`note_sheet` G12)
the coupling is mutual			
B9	the tissue feels the matrix: shape differs from the free control	any separation	**one-way** — the reaction is computed, recorded and dropped
B10	the growth gate changes the shape it is given	oblateness against the unconfined run	**1.43** grown vs **1.41** pressed (`58`, `48`)

One convention makes these mean what they say. A reaction is measured on its own force pair, inside the substep and before any drive is added; with the drive in the sum it stops being a statement about the reaction. That is why B3 and B4 are separate rows with different thresholds: B3 is float32 through a grid, B4 is float64 on an explicit edge.

What the gates license. The epithelium is reproducible on its own and nothing attached to it has moved it (B1, B2); every arrow that has been built returns its reaction to machine precision (B3, B4); the sheet, as a surface, reports the deformation it is given (B6); and the growth gate can produce a shape that pressing produces too, which is the one place the matrix has been shown to change the tissue (B10). **What they do not license** is the assembled model: B5 is not built, B7 and B8 fail on the sheet's own testbench, and B9 is the architecture — the tissue is a replay, so it drives and is never driven. Until B9 passes, the stroma's modulus is very nearly a free parameter: the tissue never feels the matrix, and the growth gate normalises the pressure map to its own late-run median, so a softer stroma changes the substep it needs and not the shape that comes out. That is the clearest statement of what one-way coupling costs, and `mesh_contact` returning `vertex_force` to a live epithelium is the run that ends it.

What sixty-six archived runs settled, and it is four things. Runs `91`–`164` asked where the sheet sits and what holds it there, and they answer that question and no other. The standoff is a *tracking lag*, $\delta - \dot{R}\Delta t\gamma/\kappa$, and not a force balance, which two fibre lengths differing fourfold confirmed to 3×10^{-5} box units. A force routed through the grid leaves the sheet carrying 100% of its true stretch while the same force integrated by the engine leaves it carrying 13% (B6). An MPM fibre cannot act as a rope, because $\mathbf{F}$ is advected by the grid's velocity gradient and never measures the distance between two particles. And every MPM body in the prototype silently carried the *parent* set's modulus, so a membrane declaring $E = 400$ ran at the stroma's 15 until each body was given its own parent — which is why run 146 came back bit-identical to run 144 at ten times the stiffness. None of the four is a model derived from what a basement membrane *is*; that is what `note_sheet` builds.

5. References

References

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